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Uniform unlikely intersections for unicritical polynomials

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Abstract. Fix $d \geq 2$, and consider the family $f_t(z) = z^d + t$ of polynomials parameterized by $t \in \mathbb{C}$. In this article, we will show that there exists a constant $C(d)$ such that for any $a, b \in \mathbb{C}$ with $a^d \neq b^d$, the number of $t \in \mathbb{C}$ such that a and b are both preperiodic for f_t is at most $C(d)$.

Keywords. Unlikely intersections, unicritical polynomials, preperiodic points, Arakelov–Zhang pairing

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1. Introduction

Fix an integer $d \geq 2$, and consider the family

$$f_t(z) = z^d + t$$

of polynomials parameterized by $t \in \mathbb{C}$. A point $a \in \mathbb{C}$ is said to be *preperiodic* for f_t if its forward orbit $O_{f_t}(a) = \{f_t^n(a) : n \geq 1\}$ is finite, where $f_t^n(a)$ is the nth iterate of a under f_t.

Let $\mathrm{PrePer}(f_t)$ be the set of all preperiodic points of f_t. By [Bea90, Section 4], [BD11, Theorem 1.2 and Corollary 1.3], and [YZ13, Theorem 1.3], the intersection $\mathrm{PrePer}(f_{t_1}) \cap \mathrm{PrePer}(f_{t_2})$ is finite if and only if $t_1 \neq t_2$. Furthermore, when $d = 2$, DeMarco, Krieger, and Ye [DKY22, Theorem 1.1] proved that $\mathrm{PrePer}(f_{t_1}) \cap \mathrm{PrePer}(f_{t_2})$ is in fact uniformly bounded for any $t_1 \neq t_2$. This result is an analog of [DKY20, Theorem 1.4], which provides a partial solution to the effective finiteness conjecture for elliptic curves proposed in [BFT18] and also implies a uniform Manin–Mumford bound under suitable conditions; see [DKY20, Theorem 1.1].

On the other hand, for any $a, b \in \mathbb{C}$, we define

$$S_{a,b} = \{t \in \mathbb{C} : a \text{ and } b \text{ are both preperiodic for } f_t\}.$$

Note that $S_{a,b}$ depends on d implicitly. Zannier [Zan12, Section 3.4.7] asked whether $S_{0,1}$ is finite when $d = 2$. Baker and DeMarco [BD11, Theorem 1.1] answered this question affirmatively by showing that $S_{a,b}$ is finite if and only if $a^d \neq b^d$. This result is motivated by a theorem of Masser and Zannier. In [MZ08, MZ10, MZ12], they showed that for any $a \neq b \in \mathbb{C}\backslash\{0, 1\}$, there exist only finitely many $t \in \mathbb{C}$ such that

$$\left(a, \sqrt{a(a-1)(a-t)}\right) \quad \text{and} \quad \left(b, \sqrt{b(b-1)(b-t)}\right)$$

are both torsion on the elliptic curve $y^2 = x(x-1)(x-t)$. For further developments on this problem, see also [BD13, GHT13, GHT15].

However, the proof of [BD11, Theorem 1.1] is not effective, so no explicit upper bound for $|S_{a,b}|$ is given. In order to find more information on $S_{a,b}$, Fili [Fil17] studied the case $S_{0,1}$ when $d = 2$ and, more importantly, made a key observation, see [Fil17, Theorem 1], which turns out to be crucial for [DKY20] and [DKY22]. In light of [DKY20, Theorem 1.4] and [DKY22, Theorem 1.1], one is led to ask whether $S_{a,b}$ is also uniformly bounded for any $a, b \in \mathbb{C}$ with $a^d \neq b^d$. In this article, we are able to show that the answer is yes.

Theorem 1.1. *For any integer $d \geq 2$, there exists a constant $C(d)$ such that $|S_{a,b}| \leq C(d)$ for any $a, b \in \mathbb{C}$ with $a^d \neq b^d$.*

Our results and proofs are inspired by [DKY22], and our techniques and strategies also come from their ideas. By a standard specialization argument, it suffices to prove Theorem 1.1 for $a, b \in \overline{\mathbb{Q}}$. Let K be a number field such that $a, b \in K$, and let M_K be the set of places. For each $v \in M_K$, we work with the dynamics of f_t on the Berkovich affine line $\mathbb{A}_v^{1,\mathrm{an}}$. In Section 2, we will introduce

- the generalized Mandelbrot sets $M_{a,v}$ and $M_{b,v}$,
- the equilibrium measures $\mu_{a,v}$ and $\mu_{b,v}$,
- the Green's functions $g_{a,v}$ and $g_{b,v}$.

Then the Arakelov–Zhang pairing of $\boldsymbol{\mu}_a = \{\mu_{a,v}\}_{v\in M_K}$ and $\boldsymbol{\mu}_b = \{\mu_{b,v}\}_{v\in M_K}$ is given by

$$\left\langle \boldsymbol{\mu}_a, \boldsymbol{\mu}_b \right\rangle = \sum_{v\in M_K} \frac{[K_v : \mathbb{Q}_v]}{[K : \mathbb{Q}]} \int_{\mathbb{A}_v^{1,\mathrm{an}}} g_{a,v}\, d\mu_{b,v}.$$

The value of $\langle \boldsymbol{\mu}_a, \boldsymbol{\mu}_b \rangle$ depends on a and b only and is independent of the choice of K. By [FRL06, Propositions 2.6 and 4.5] and [BD11, Theorems 1.1 and 3.4], we have $\langle \boldsymbol{\mu}_a, \boldsymbol{\mu}_b \rangle = \langle \boldsymbol{\mu}_b, \boldsymbol{\mu}_a \rangle \geq 0$ and

$$\left\langle \boldsymbol{\mu}_a, \boldsymbol{\mu}_b \right\rangle = 0 \iff \boldsymbol{\mu}_a = \boldsymbol{\mu}_b \iff a^d = b^d \iff |S_{a,b}| = \infty.$$

Given this equivalence relation, it is not surprising that the value of $\langle \boldsymbol{\mu}_a, \boldsymbol{\mu}_b \rangle$ encodes some information about $S_{a,b}$. As in [DKY20, DKY22], the main task of this article is to estimate the upper and lower bounds for $\langle \boldsymbol{\mu}_a, \boldsymbol{\mu}_b \rangle$. More precisely, we prove the following counterparts of [DKY20, Theorems 1.5, 1.6, and 1.7] and [DKY22, Theorems 1.6, 1.7, and 1.9].

Theorem 1.2. *Let $a, b \in \overline{\mathbb{Q}}$ such that $a^d \neq b^d$ and $|S_{a,b}| > 0$. For any $0 < \varepsilon < 4d$, we have*

$$\left\langle \boldsymbol{\mu}_a, \boldsymbol{\mu}_b \right\rangle \leq \left(\varepsilon + \frac{8d/\varepsilon - 2}{|S_{a,b}|} \right) (h(a,b) + 5),$$

where h is the logarithmic Weil height on $\mathbb{A}^2(\overline{\mathbb{Q}})$.

Theorem 1.3. *For any $a, b \in \overline{\mathbb{Q}}$ such that $a^d \neq b^d$, we have*

$$\left\langle \boldsymbol{\mu}_a, \boldsymbol{\mu}_b \right\rangle \geq \frac{1}{12d^2} h(a,b) - 1,$$

where h is the logarithmic Weil height on $\mathbb{A}^2(\overline{\mathbb{Q}})$.

Theorem 1.4. *There exists a constant $\delta(d) > 0$ such that $\langle \boldsymbol{\mu}_a, \boldsymbol{\mu}_b \rangle \geq \delta(d)$ for any $a, b \in \overline{\mathbb{Q}}$ with $a^d \neq b^d$.*

Note that Theorem 1.3 is trivial when $h(a,b)$ is small. We need Theorem 1.4 to deal with this case.

The main differences between this article and [DKY22] are: (1) Our Theorem 1.1 is valid for any $d \geq 2$, while [DKY22, Theorem 1.1] focuses on $d = 2$. (2) We work with the generalized Mandelbrot sets instead of the Julia sets in [DKY22]. (3) When we estimate the lower bounds for $\langle \mu_{a,v}, \mu_{b,v} \rangle_v$ at the non-Archimedean places $v \in M_K^0$, our computations in Section 4.4 are more simplified than those in [DKY22, Sections 5 and 6]. This simplification helps us to work with all $d \geq 2$ at the same time.

The plan of this article is as follows: In Section 2, we fix the notation and review the tools we will need. In Section 3, we first estimate the upper bounds for $\langle \boldsymbol{\mu}_a, \boldsymbol{\mu}_b \rangle$ locally, and then we combine the local estimates to give the proof of Theorem 1.2. The structure of Section 4 is similar, but this time we estimate the lower bounds for $\langle \boldsymbol{\mu}_a, \boldsymbol{\mu}_b \rangle$ and give the proofs of Theorems 1.3 and 1.4. In Section 5, we show that Theorem 1.1 follows from Theorems 1.2, 1.3, and 1.4.

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2. Notation and preliminaries

The main references for this section are [BR10, BD11, FRL06, Fil17].

Given a number field K, let M_K be the set of places, let M_K^∞ be the set of Archimedean places, and let M_K^0 be the set of non-Archimedean places. We normalize the absolute values $|\cdot|_v$ on K such that they extend the standard absolute values on $\mathbb{Q}$. For any $a \in K^\times$, we have the product formula

$$\prod_{v \in M_K} |a|_v^{n_v} = 1, \quad \text{where } n_v = \frac{[K_v : \mathbb{Q}_v]}{[K : \mathbb{Q}]}.$$

2.1. Berkovich spaces

For each $v \in M_K$, let K_v be the completion of K at v, let $\overline{K}_v$ be an algebraic closure of K_v, and let $\mathbb{C}_v$ be the completion of $\overline{K}_v$. The *Berkovich affine line* $\mathbb{A}_v^{1,\mathrm{an}}$ is a locally compact, Hausdorff, path-connected space containing $\mathbb{C}_v$ as a dense subspace. As a topological space, $\mathbb{A}_v^{1,\mathrm{an}}$ is the set of all multiplicative seminorms $[\cdot]_x \colon \mathbb{C}_v[T] \to \mathbb{R}$ on the polynomial ring $\mathbb{C}_v[T]$ that extend the absolute value $|\cdot|_v$ on $\mathbb{C}_v$, endowed with the weakest topology for which $x \mapsto [f]_x$ is continuous for any $f \in \mathbb{C}_v[T]$. The *Berkovich projective line* $\mathbb{P}_v^{1,\mathrm{an}}$ can be identified with the one-point compactification of $\mathbb{A}_v^{1,\mathrm{an}}$.

If $v \in M_K^\infty$, then by the Gelfand–Mazur theorem, $\mathbb{A}_v^{1,\mathrm{an}}$ is homeomorphic to $\mathbb{C}_v = \mathbb{C}$. If $v \in M_K^0$, then by Berkovich's classification theorem, each $x \in \mathbb{A}_v^{1,\mathrm{an}}$ corresponds to a decreasing nested sequence $\{\overline{D}(a_n, r_n)\}_{n=1}^\infty$ of closed disks on $\mathbb{C}_v$ such that

$$[f]_x = \lim_{n \to \infty} \sup_{z \in \overline{D}(a_n, r_n)} |f(z)|_v.$$

Based on the nature of $D = \cap_{n=1}^\infty \overline{D}(a_n, r_n)$, the points of $\mathbb{A}_v^{1,\mathrm{an}}$ can be categorized into four types: (I) D is a point of $\mathbb{C}_v$, (II) D is a closed disk with radius in $|\mathbb{C}_v^\times|$, (III) D is a closed disk with radius not in $|\mathbb{C}_v^\times|$, and (IV) D is the empty set.

For any $a \in \mathbb{C}_v$ and any $r > 0$, we define $\mathcal{D}(a, r)$ to be the set of points corresponding to $\{\overline{D}(a_n, r_n)\}_{n=1}^\infty$ with $\overline{D}(a_n, r_n) \subseteq \overline{D}(a, r)$ and define $\zeta_{a,r}$ to be the point corresponding to $\overline{D}(a, r)$.

2.2. Potential theory

When $v \in M_K^0$, we introduce the *Hsia kernel*

$$\delta_v(x, y) = \limsup_{z, w \in \mathbb{C}_v, z \to x, w \to y} |z - w|_v,$$

which extends the distance function $|x - y|_v$ on $\mathbb{C}_v$ to the entire $\mathbb{A}_v^{1,\mathrm{an}}$. When $v \in M_K^\infty$, we also write $\delta_v(x, y) = |x - y|_v$ to unify the notation further on.

Fix a $v \in M_K$, and let E be a compact subset of $\mathbb{A}_v^{1,\mathrm{an}}$. The *logarithmic capacity* $\gamma_v(E)$ of E is given by

$$-\log \gamma_v(E) = \inf_{\mu} \iint_{E \times E} -\log \delta_v(z, w) d\mu(z) d\mu(w),$$

where the infimum is taken over all probability measures μ supported on E. If $\gamma_v(E) > 0$, then there exists a unique probability measure μ_E, called the *equilibrium measure* of E, such that the infimum is achieved. The *Green's function* of E is defined by

$$g_E(z) = -\log \gamma_v(E) + \int_E \log \delta_v(z, w) d\mu_E(w),$$

which is a non-negative real-valued function on $\mathbb{A}_v^{1,\mathrm{an}}$.

2.3. Generalized Mandelbrot sets

Now we go back to the dynamics of $f_t(z) = z^d + t$. For any $v \in M_K$ and any $a \in \mathbb{C}_v$, we define the *generalized Mandelbrot set* by

$$M_{a,v} = \left\{ t \in \mathbb{A}_v^{1,\mathrm{an}} : \sup_n \left[f_T^n(a) \right]_t < \infty \right\},$$

where $f_T^n(a)$ is considered as an element of the polynomial ring $\mathbb{C}_v[T]$. Note that if $t \in \mathbb{C}_v$, then $[f_T^n(a)]_t$ is simply $|f_t^n(a)|_v$. It is known that $M_{a,v}$ is compact. We write $\mu_{a,v}$ and $g_{a,v}$ for the equilibrium measure and the Green's function of $M_{a,v}$. The following properties are collected from [BD11, Section 3].

Theorem 2.1. *For any $v \in M_K$ and any $a \in \mathbb{C}_v$, we have the following:*

(1) *The logarithmic capacity $\gamma_v(M_{a,v})$ equals 1.*
(2) *The Green's function of $M_{a,v}$ is given by*

$$g_{a,v}(t) = \lim_{n\to\infty} \frac{1}{d^n} \log^+ \left[f_T^{n+1}(a) \right]_t,$$

where $\log^+ z = \log \max\{z, 1\}$ for any $z \in \mathbb{R}$.
(3) *The function $g_{a,v}(t)$ is continuous on $\mathbb{A}_v^{1,\mathrm{an}}$.*
(4) *The function $g_{a,v}(t)$ is harmonic on $\mathbb{A}_v^{1,\mathrm{an}} \backslash M_{a,v}$.*
(5) *We have $g_{a,v}(t) = 0$ if and only if $t \in M_{a,v}$.*

2.4. Arakelov–Zhang pairing

Like [DKY20, Theorem 1.7] and [DKY22, Theorem 1.9], our Theorem 1.2 also builds on the quantitative equidistribution results of [FRL06, Fil17].

Let $a \in \mathbb{C}_v$ and $r > 0$. If $v \in M_K^\infty$, we define $m_{a,r,v}$ to be the normalized Haar measure on the circle $\partial D(a, r)$. If $v \in M_K^0$, we define $m_{a,r,v}$ to be the Dirac measure on $\zeta_{a,r}$.

Definition 2.2 (*cf.* [FRL06, Définition 1.1]). We call $\boldsymbol{\mu} = (\mu_v)_{v\in M_K}$ an *adelic measure* if

(1) μ_v is a probability measure on $\mathbb{P}_v^{1,\mathrm{an}}$ for any $v \in M_K$,
(2) $\mu_v = m_{0,1,v}$ for all but finitely many $v \in M_K$,
(3) for any $v \in M_K$, we have $\mu_v - m_{0,1,v} = \Delta u_v$ for some continuous function u_v on $\mathbb{P}_v^{1,\mathrm{an}}$, where Δ is the Laplacian on $\mathbb{P}_v^{1,\mathrm{an}}$.

Following [FRL06, Sections 2.4 and 4.4], for each $v \in M_K$, we define the *mutual energy* of two signed measures $\mu_{1,v}$ and $\mu_{2,v}$ on $\mathbb{P}_v^{1,\mathrm{an}}$ by

$$(\mu_{1,v}, \mu_{2,v})_v = \iint_{\mathbb{A}_v^{1,\mathrm{an}} \times \mathbb{A}_v^{1,\mathrm{an}} \backslash \mathrm{Diag}_v} -\log \delta_v(z, w) d\mu_{1,v}(z) d\mu_{2,v}(w),$$

where Diag_v is the diagonal on $\mathbb{C}_v \times \mathbb{C}_v$. Suppose $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_2$ are adelic measures; then we define their *v-adic Arakelov–Zhang pairing* by

$$\langle \mu_{1,v}, \mu_{2,v} \rangle_v = \frac{1}{2} (\mu_{1,v} - \mu_{2,v}, \mu_{1,v} - \mu_{2,v})_v$$

and define their *Arakelov–Zhang pairing* by

$$\left\langle \boldsymbol{\mu}_1, \boldsymbol{\mu}_2 \right\rangle = \sum_{v\in M_K} n_v \langle \mu_{1,v}, \mu_{2,v} \rangle_v.$$

Theorem 2.3 (*cf.* [Fil17, Theorem 1]). *The square root of the Arakelov–Zhang pairing $\langle \cdot, \cdot \rangle^{1/2}$ gives a metric on the space of all adelic measures.*

Now we go back to the dynamics of $f_t(z) = z^d + t$. Given $a, b \in K$, let $\boldsymbol{\mu}_a = \{\mu_{a,v}\}_{v\in M_K}$ and $\boldsymbol{\mu}_b = \{\mu_{b,v}\}_{v\in M_K}$ be the equilibrium measures defined in Section 2.3. It is known that they are adelic measures, and their

v-adic Arakelov–Zhang pairing can be written as

$$\langle \mu_{a,v}, \mu_{b,v} \rangle_v = \int_{\mathbb{A}_v^{1,\mathrm{an}}} g_{a,v} d\mu_{b,v}.$$

For the Arakelov–Zhang pairing in more general settings, see [Zha95, PST12, CL11].

3. Upper bounds

The purpose of this section is to give the proof of Theorem 1.2. In order to do so, we estimate the upper bounds for $g_{a,v}(s)$ when s is close to $M_{a,v}$. We work with $v \in M_K^\infty$ in Section 3.1 and $v \in M_K^0$ in Section 3.2. In Section 3.3, we apply the local estimates in Theorem 3.13 to complete the proof of Theorem 1.2.

3.1. Archimedean estimates

In this section, we assume that K is a number field, $a \in K$, and $v \in M_K^\infty$. Because v is fixed, we write $|\cdot|$, M_a, and g_a for $|\cdot|_v$, $M_{a,v}$, and $g_{a,v}$.

From [BD11, Lemma 3.2], we know that M_a is bounded. Their proof can be modified slightly to give an explicit bound for M_a. We begin with a basic distortion result for univalent maps, which is used in the proof of [BD11, Lemma 3.2] and also in [DKY22, Section 3.1].

Theorem 3.1. *Let $U_R = \{z \in \mathbb{C} : |z| > R\}$. If $\phi\colon U_R \to \mathbb{C}$ is analytic and injective, and*

$$\phi(z) = z + \sum_{n=1}^{\infty} \frac{a_n}{z^n},$$

then $\phi(U_R) \supseteq U_{2R}$. In particular, $|\phi(z)| \le 2|z|$ for any $z \in U_R$.

Proof. The first assertion is [BH88, Corollary 3.3]. If the second assertion is false, then there exists a $z \in U_R$ such that $\phi(z) \in U_{2|z|} \subseteq \phi(U_{|z|})$, which contradicts the injectivity of ϕ. □

Proposition 3.2. *If $|t| > 4^d \max\{|a|, 4\}^d$, then*

$$\log|t| - 1 \le g_a(t) \le \log|t| + 1.$$

In particular, if $t \in M_a$, then $|t| \le 4^d \max\{|a|, 4\}^d$.

Proof. For each $t \in \mathbb{C}$, let

$$\lambda_t(z) = \lim_{n\to\infty} \frac{1}{d^n} \log^+ |f_t^n(z)|.$$

There exists an analytic homeomorphism ϕ_t, defining the *Böttcher coordinate* near ∞, which satisfies $\phi_t(f_t(z)) = \phi_t(z)^d$ and $\lambda_t(z) = \log|\phi_t(z)|$. The map ϕ_t sends the domain

$$V_t = \{z \in \mathbb{C} : \lambda_t(z) > \lambda_t(0)\}$$

biholomorphically to U_{R_t} with $R_t = e^{\lambda_t(0)}$. By Theorem 3.1, we have $V_t = \phi_t^{-1}(U_{R_t}) \supseteq U_{2R_t}$. By the proof of [BD11, Lemma 3.2], we have $|t| \le 2^d R_t^d$, and if t is large enough such that $R_t^d - 4R_t > 2|a|^d$, then $a^d + t \in U_{2R_t}$. If $|t| > 4^d \max\{|a|, 4\}^d$, then we have

$$R_t \ge \frac{1}{2}|t|^{1/d} > 2\max\{|a|, 4\}$$

and

$$\begin{aligned} R_t^d - 4R_t &= R_t\left(R_t^{d-1} - 4\right) > 2\max\{|a|, 4\}\left(2^{d-1}\max\{|a|, 4\}^{d-1} - 4\right) \\ &= 2\max\{|a|, 4\}^d + \left(\left(2^d - 2\right)\max\{|a|, 4\}^{d-1} - 8\right)\max\{|a|, 4\} \ge 2|a|^d. \end{aligned}$$

Applying Theorem 3.1 to ϕ_t, we get

$$\left|\phi_t\left(a^d+t\right)\right| \le 2\left|a^d+t\right| \le 2\left(|a|^d+|t|\right) \le \frac{17}{8}|t|,$$

where the last inequality follows from $|t| > 4^d \max\{|a|,4\}^d \ge 16|a|^d$. Applying Theorem 3.1 to ϕ_t^{-1}, we get

$$\left|\phi_t\left(a^d+t\right)\right| \ge \frac{1}{2}\left|\phi_t^{-1}\left(\phi_t\left(a^d+t\right)\right)\right| = \frac{1}{2}\left|a^d+t\right| \ge \frac{1}{2}\left(|t|-|a|^d\right) \ge \frac{15}{32}|t|.$$

By [BD11, Proposition 3.3], we have

$$\log|t|-1 \le g_a(t) = \log\left|\phi_t\left(a^d+t\right)\right| \le \log|t|+1.$$

In particular, by Theorem 2.1, if $|t| > 4^d\max\{|a|,4\}^d$, then $g_a(t) > 0$ and $t \notin M_a$. □

By definition, $O_{f_t}(a)$ is bounded if $t \in M_a$. The following result shows that $O_{f_t}(a)$ is in fact uniformly bounded for any $t \in M_a$.

Proposition 3.3. *If $t \in M_a$, then $|f_t^n(a)| < 8\max\{|a|,4\}$ for any $n \ge 1$.*

Proof. By Proposition 3.2, if $t \in M_a$, then $|t| \le 4^d\max\{|a|,4\}^d$. Suppose there exists an $n \ge 1$ such that $|f_t^n(a)| \ge 8\max\{|a|,4\}$; then

$$\left|f_t^{n+1}(a)\right| \ge |f_t^n(a)|^d - |t| \ge \left(1-\frac{1}{2^d}\right)|f_t^n(a)|^d \ge 24|f_t^n(a)|.$$

By induction, $|f_t^{n+k}(a)| \ge 24^k|f_t^n(a)| \to \infty$ as $k \to \infty$, and we have a contradiction. □

The explicit bound of M_a given in Proposition 3.2 can be improved as follows.

Proposition 3.4. *If $t \in M_a$, then $|t| < 3\max\{|a|,4\}^d$.*

Proof. If $t \in M_a$, then by Proposition 3.3,

$$\begin{aligned}|t| &\le \left|a^d+t\right| + |a|^d < 8\max\{|a|,4\} + |a|^d \\ &\le \left(\frac{8}{\max\{|a|,4\}^{d-1}}+1\right)\max\{|a|,4\}^d \le 3\max\{|a|,4\}^d.\end{aligned}$$

□

As [DKY22, Section 3.3], we estimate the upper bound for $g_a(s)$ when s is close to M_a.

Proposition 3.5. *If $t \in M_a$ and $|s-t| \le \max\{|a|,4\}$, then for any $n \ge 1$,*

$$|f_s^n(a) - f_t^n(a)| \le |s-t|(18\max\{|a|,4\})^{d^{n-1}-1}.$$

Proof. Let $A_n = (18\max\{|a|,4\})^{d^{n-1}-1}$ for any $n \ge 1$. We prove the assertion by induction. The statement is true for $n=1$ because $|f_s(a)-f_t(a)| = |s-t|$. Assume the statement is true for some $n \ge 1$; then

$$\begin{aligned}\left|f_s^{n+1}(a) - f_t^{n+1}(a)\right| &= \left|f_s^n(a)^d + s - f_t^n(a)^d - t\right| \\ &\le \left|f_s^n(a)^d - f_t^n(a)^d\right| + |s-t| \\ &= \textstyle\prod_{i=0}^{d-1}\left|f_s^n(a) - f_t^n(a) + \left(1-\zeta_d^i\right)f_t^n(a)\right| + |s-t| \\ &\le |f_s^n(a)-f_t^n(a)|(|f_s^n(a)-f_t^n(a)| + 2|f_t^n(a)|)^{d-1} + |s-t| \\ &\le |s-t|A_n(A_n\max\{|a|,4\} + 16\max\{|a|,4\})^{d-1} + |s-t| \\ &\quad \text{(by the assumption, the induction hypothesis, and Proposition 3.3)} \\ &= |s-t|\left(A_n(A_n+16)^{d-1}\max\{|a|,4\}^{d-1}+1\right) \\ &\le |s-t|\left(A_n(A_n+16A_n)^{d-1}\max\{|a|,4\}^{d-1} + \max\{|a|,4\}^{d-1}\right)\end{aligned}$$

$$\begin{aligned}&= |s-t|\left(17^{d-1}A_n^d + 1\right)\max\{|a|,4\}^{d-1}\\ &\le |s-t|\,18^{d-1}A_n^d \max\{|a|,4\}^{d-1}\\ &= |s-t|A_{n+1}.\end{aligned}$$

This completes the inductive step and hence the proof. □

Proposition 3.6. *If $t \in M_a$ and*

$$|s-t| \le \frac{1}{18}(18\max\{|a|,4\})^{2-d^{n-1}}$$

for some $n \ge 1$, then

$$g_a(s) \le \frac{1}{d^{n-1}}\log(10\max\{|a|,4\}).$$

Proof. Let $A_n = (18\max\{|a|,4\})^{d^{n-1}-1}$ for any $n \ge 1$. By Propositions 3.3 and 3.5,

$$|f_s^n(a)| \le |f_s^n(a) - f_t^n(a)| + |f_t^n(a)| \le |s-t|A_n + |f_t^n(a)| \le 9\max\{|a|,4\}.$$

By Proposition 3.4,

$$\left|f_s^{n+1}(a)\right| \le |f_s^n(a)|^d + |t| + |s-t| \le \left(9^d + 4\right)\max\{|a|,4\}^d.$$

Let $p(z) = z^d + 4$; then by induction and Lemma 3.7,

$$\left|f_s^{n+k}(a)\right| \le p^k(9)\max\{|a|,4\}^{d^k} \le (10\max\{|a|,4\})^{d^k}$$

for any $k \ge 1$. Therefore, by Theorem 2.1,

$$g_a(s) = \lim_{k\to\infty}\frac{1}{d^{n+k-1}}\log^+\left|f_s^{n+k}(a)\right| \le \frac{1}{d^{n-1}}\log(10\max\{|a|,4\}).$$ □

Lemma 3.7. *Let $p(z) = z^d + 4$ for some $d \ge 2$. Then $p^n(9) \le 10^{d^n}$ for any $n \ge 1$.*

Proof. We prove $q(n) = 10^{d^n} - p^n(9) \ge 1$ for any $n \ge 0$ by induction. It is clear that $q(0) = 1$. Assume $q(n) \ge 1$ for some $n \ge 0$; then

$$\begin{aligned}q(n+1) &= \left(10^{d^n}\right)^d - p^n(9)^d - 4 = q(n)\textstyle\sum_{i=0}^{d-1}\left(10^{d^n}\right)^i p^n(9)^{d-1-i} - 4\\ &\ge q(n)10^{d^n(d-1)} - 4 \ge 1.\end{aligned}$$

This completes the inductive step and hence the proof. □

3.2. Non-Archimedean estimates

In this section, we assume that K is a number field, $a \in K$, and $v \in M_K^0$. Because v is fixed, we write $|\cdot|$, δ, M_a, and g_a for $|\cdot|_v$, δ_v, $M_{a,v}$, and $g_{a,v}$.

Propositions 3.8 and 3.9 can be seen as the non-Archimedean versions of Propositions 3.2, 3.3, and 3.4.

Proposition 3.8. *Assume that $|a| \le 1$. For any $t \in \mathbb{C}_v$, we have*

$$g_a(t) = \log^+|t|.$$

In particular, $M_a = \mathcal{D}(0,1)$, the closed Berkovich unit disk.

Proof. Let $t \in \mathbb{C}_v$; then for any $n \ge 1$,

$$|f_t^n(a)|\begin{cases}\le 1 & \text{if } |t| \le 1,\\ = |t|^{d^{n-1}} & \text{if } |t| > 1.\end{cases}$$

By Theorem 2.1,

$$g_a(t) = \lim_{n\to\infty}\frac{1}{d^{n-1}}\log^+|f_t^n(a)| = \log^+|t|.$$

Therefore, $M_a \cap \mathbb{C}_v = \overline{D}(0,1)$ and $M_a = \overline{M_a \cap \mathbb{C}_v} = \mathcal{D}(0,1)$. □

Proposition 3.9. *Assume that* $|a| > 1$. *For any* $t \in \mathbb{C}_v$, *we have*

$$g_a(t)\begin{cases} = d\log|a| & \text{if } |t| < |a|^d, \\ \le d\log|a| & \text{if } |t| = |a|^d, \\ = \log|t| & \text{if } |t| > |a|^d. \end{cases}$$

If $t \in M_a \cap \mathbb{C}_v$, *then* $|t| = |a|^d$ *and* $|f_t^n(a)| = |a|$ *for any* $n \ge 1$.

Proof. Let $t \in \mathbb{C}_v$; then for any $n \ge 1$,

$$|f_t^n(a)|\begin{cases} = |a|^{d^n} & \text{if } |t| < |a|^d, \\ \le |a|^{d^n} & \text{if } |t| = |a|^d, \\ = |t|^{d^{n-1}} & \text{if } |t| > |a|^d. \end{cases}$$

By Theorem 2.1,

$$g_a(t) = \lim_{n\to\infty} \frac{1}{d^{n-1}} \log^+ |f_t^n(a)|\begin{cases} = d\log|a| & \text{if } |t| < |a|^d, \\ \le d\log|a| & \text{if } |t| = |a|^d, \\ = \log|t| & \text{if } |t| > |a|^d. \end{cases}$$

Therefore, if $t \in M_a \cap \mathbb{C}_v$, then $|t| = |a|^d$ and $t \in M_{f_t^n(a)} \cap \mathbb{C}_v$ for any $n \ge 1$. By the same reasoning, we have $|t| = |f_t^n(a)|^d$ and $|f_t^n(a)| = |a|$. □

The following result will not be used until Section 4. We include it here because it is similar to Proposition 3.9.

Proposition 3.10. *If* $|a| > 1$, $t \in \mathbb{C}_v$, *and* $|f_t^n(a)| < |a|^d$ *for some* $n \ge 1$, *then* $|t| = |a|^d$ *and* $|f_t^k(a)| = |a|$ *for any* $1 \le k \le n-1$.

Proof. For the first assertion, suppose $|t| \ne |a|^d$; then

$$|f_t^n(a)| = \max\left\{|t|, |a|^d\right\}^{d^{n-1}} \ge |a|^d.$$

For the second assertion, suppose $|f_t^k(a)| \ne |a|$ for some $1 \le k \le n-1$; then

$$|f_t^n(a)| = \max\left\{|t|, \left|f_t^k(a)\right|^d\right\}^{d^{n-k-1}} \ge |t| = |a|^d.$$ □

Propositions 3.11 and 3.12 can be seen as the non-Archimedean versions of Propositions 3.5 and 3.6.

Proposition 3.11. *If* $|a| > 1$, $t \in M_a \cap \mathbb{C}_v$, $s \in \mathbb{C}_v$, *and* $|s-t| \le |a|$, *then for any* $n \ge 1$,

$$|f_s^n(a) - f_t^n(a)| \le |s-t| \cdot |a|^{d^{n-1}-1}.$$

Proof. We prove the assertion by induction. Because $|f_s(a) - f_t(a)| = |s-t|$, the statement is true for $n = 1$. Assume the statement is true for some $n \ge 1$; then

$$\begin{aligned} \left|f_s^{n+1}(a) - f_t^{n+1}(a)\right| &= \left|f_s^n(a)^d + s - f_t^n(a)^d - t\right| \\ &\le \max\left\{\left|f_s^n(a)^d - f_t^n(a)^d\right|, |s-t|\right\} \\ &= \max\left\{\prod_{i=0}^{d-1}\left|f_s^n(a) - f_t^n(a) + (1-\zeta_d^i)f_t^n(a)\right|, |s-t|\right\} \\ &\le \max\left\{|f_s^n(a) - f_t^n(a)|\max\{|f_s^n(a) - f_t^n(a)|, |f_t^n(a)|\}^{d-1}, |s-t|\right\} \\ &\le \max\left\{|s-t|\cdot|a|^{d^{n-1}-1}\max\left\{|a|^{d^{n-1}}, |a|\right\}^{d-1}, |s-t|\right\} \end{aligned}$$

(by the assumption, the induction hypothesis, and Proposition 3.9)

$$= |s-t| \cdot |a|^{d^n-1}.$$

This completes the inductive step and hence the proof. □

Proposition 3.12. *If $t \in M_a \cap \mathbb{C}_v$, $s \in \mathbb{A}_v^{1,\mathrm{an}}$, and*

$$\delta(s,t) \le \max\{|a|,1\}^{2-d^{n-1}}$$

for some $n \ge 1$, then

$$g_a(s) \le \frac{1}{d^{n-1}} \log^+ |a|.$$

Proof. Let $A_n = \max\{|a|,1\}^{2-d^{n-1}}$ for any $n \ge 1$. By Theorem 2.1, the function g_a is continuous. Since $\overline{D}(t,A_n)$ is dense in $\mathcal{D}(t,A_n)$, it suffices to prove the assertion for $s \in \overline{D}(t,A_n)$. If $|a| \le 1$, then by Proposition 3.8, we have $s \in M_a$ and $g_a(s) = 0$. If $|a| > 1$, then by Propositions 3.9 and 3.11,

$$|f_s^n(a)| \le \max\{|f_s^n(a) - f_t^n(a)|, |f_t^n(a)|\} \le |a|$$

and

$$\left|f_s^{n+1}(a)\right| \le \max\left\{|f_s^n(a)|^d, |t|, |s-t|\right\} \le |a|^d.$$

By induction, $|f_s^{n+k}(a)| \le |a|^{d^k}$ for any $k \ge 1$. Therefore, by Theorem 2.1,

$$g_a(s) = \lim_{k\to\infty} \frac{1}{d^{n+k-1}} \log^+ \left|f_s^{n+k}(a)\right| \le \frac{1}{d^{n-1}} \log |a|.$$

□

3.3. Proof of Theorem 1.2

Now we are ready to give the proof of Theorem 1.2. Given $a, b \in \overline{\mathbb{Q}}$, let K be a number field such that $a, b \in K$. As in [DKY22, Section 9], we will apply [Fil17, Theorem 1] in the following way:

$$\left\langle \boldsymbol{\mu}_a, \boldsymbol{\mu}_b \right\rangle^{1/2} \le \left\langle \boldsymbol{\mu}_a, [S]_{\boldsymbol{\tau}} \right\rangle^{1/2} + \left\langle \boldsymbol{\mu}_b, [S]_{\boldsymbol{\tau}} \right\rangle^{1/2},$$

where $[S]_{\boldsymbol{\tau}}$ is an adelic measure to be described below.

Let S be a finite, non-empty, $\mathrm{Gal}(\overline{K}/K)$-invariant subset of $\overline{K}$, and let $[S]$ be the probability measure supported equally on the elements of S. We call $\boldsymbol{\tau} = \{\tau_v\}_{v \in M_K}$ an *adelic radius* if $\tau_v > 0$ for any $v \in M_K$ and $\tau_v = 1$ for all but finitely many $v \in M_K$. Combining the ideas of [FRL06, Section 4.6] and [FP19, Section 2.1], we define the *regularization* $[S]_{\boldsymbol{\tau}} = \{[S]_{\tau_v}\}_{v \in M_K}$ by

$$[S]_{\tau_v} = \frac{1}{|S|} \sum_{s \in S} m_{s,\tau_v,v},$$

where $m_{s,\tau_v,v}$ is defined in Section 2.4.

The following result and the proof of Theorem 1.2 are adapted from [DKY22, Section 9].

Theorem 3.13. *Let K be a number field such that $a, b \in K$. Then*

$$\left\langle \boldsymbol{\mu}_a, \boldsymbol{\mu}_b \right\rangle^{1/2} \le \sum_{i=a,b} \left(\sum_{v \in M_K} n_v \left(-(\mu_{i,v}, [S]_{\tau_v})_v - \frac{\log \tau_v}{2|S|} \right) \right)^{1/2}$$

for any finite, non-empty, $\mathrm{Gal}(\overline{K}/K)$-invariant subset S of $\overline{K}$ and any adelic radius $\boldsymbol{\tau}$.

Proof. By Theorem 2.1, we have $(\mu_{i,v}, \mu_{i,v})_v = -\log \gamma_v(M_{i,v}) = 0$ for $i = a, b$ and any $v \in M_K$. Then the proof is identical to that of [DKY22, Lemma 9.2]. □

Proof of Theorem 1.2. Let K be a number field such that $a, b \in K$, and let $n \geq -1$ be the integer such that $d^{-n-1} \leq \varepsilon' = \varepsilon/4 < d^{-n}$. For each $v \in M_K^\infty$, take

$$\begin{aligned}\tau_v &= (18\max\{|a|_v, |b|_v, 4\})^{1-d/\varepsilon'} \leq (18\max\{|a|_v, 4\})^{1-d/\varepsilon'} \\ &\leq \frac{1}{18}(18\max\{|a|_v, 4\})^{2-d/\varepsilon'} \leq \frac{1}{18}(18\max\{|a|_v, 4\})^{2-d^{n+1}}.\end{aligned}$$

By Proposition 3.6, if $|s-t|_v = \tau_v$ for some $t \in S_{a,b}$, then

$$g_{a,v}(s) \leq \frac{1}{d^{n+1}}\log(10\max\{|a|_v, 4\}) \leq \varepsilon' \log(18\max\{|a|_v, |b|_v, 4\}).$$

Therefore,

$$\begin{aligned}-\left(\mu_{a,v}, [S_{a,b}]_{\tau_v}\right)_v - \frac{\log\tau_v}{2|S_{a,b}|} &= \int g_{a,v}\, d[S_{a,b}]_{\tau_v} - \frac{\log\tau_v}{2|S_{a,b}|} \\ &\leq \varepsilon' \log(18\max\{|a|_v, |b|_v, 4\}) - \frac{(1-d/\varepsilon')\log(18\max\{|a|_v, |b|_v, 4\})}{2|S_{a,b}|} \\ &\leq \left(\varepsilon' + \frac{d/\varepsilon'-1}{2|S_{a,b}|}\right)\log(18\max\{4|a|_v, 4|b|_v, 4\}) \\ &\leq \left(\varepsilon' + \frac{d/\varepsilon'-1}{2|S_{a,b}|}\right)(\log\max\{|a|_v, |b|_v, 1\} + 5).\end{aligned}$$

For each $v \in M_K^0$, take

$$\begin{aligned}\tau_v &= \max\{|a|_v, |b|_v, 1\}^{1-d/\varepsilon'} \leq \max\{|a|_v, 1\}^{1-d/\varepsilon'} \\ &\leq \max\{|a|_v, 1\}^{2-d/\varepsilon'} \leq \max\{|a|_v, 1\}^{2-d^{n+1}}.\end{aligned}$$

By Proposition 3.12, if $\delta_v(s,t) = \tau_v$ for some $t \in S_{a,b}$, then

$$g_{a,v}(s) \leq \frac{1}{d^{n+1}}\log^+|a|_v \leq \varepsilon' \log\max\{|a|_v, |b|_v, 1\}.$$

Therefore,

$$\begin{aligned}-\left(\mu_{a,v}, [S_{a,b}]_{\tau_v}\right)_v - \frac{\log\tau_v}{2|S_{a,b}|} &= \int g_{a,v}\, d[S_{a,b}]_{\tau_v} - \frac{\log\tau_v}{2|S_{a,b}|} \\ &\leq \varepsilon' \log\max\{|a|_v, |b|_v, 1\} - \frac{(1-d/\varepsilon')\log\max\{|a|_v, |b|_v, 1\}}{2|S_{a,b}|} \\ &= \left(\varepsilon' + \frac{d/\varepsilon'-1}{2|S_{a,b}|}\right)\log\max\{|a|_v, |b|_v, 1\}.\end{aligned}$$

Summing over all $v \in M_K$, we get

$$\left\langle \boldsymbol{\mu}_a, \boldsymbol{\mu}_b \right\rangle \leq 4\left(\varepsilon' + \frac{d/\varepsilon'-1}{2|S_{a,b}|}\right)(h(a,b)+5) = \left(\varepsilon + \frac{8d/\varepsilon - 2}{|S_{a,b}|}\right)(h(a,b)+5). \qquad \square$$

4. Lower bounds

The purpose of this section is to give the proofs of Theorems 1.3 and 1.4. In order to do so, we estimate the lower bounds for

$$\langle \mu_{a,v}, \mu_{b,v} \rangle_v = \int_{\mathbb{A}_v^{1,\mathrm{an}}} g_{a,v}\, d\mu_{b,v}.$$

In Section 4.1, we review two equidistribution theorems for later usage. As always, we work with $v \in M_K^\infty$ and $v \in M_K^0$ separately. For some technical reasons, we also work with $d = 2$ and $d > 2$ separately when

$v \in M_K^\infty$. The local estimates obtained in Sections 4.2, 4.3, and 4.4 are gathered in Section 4.5 to complete the proofs of Theorems 1.3 and 1.4.

4.1. Equidistribution theorems

In this section, we assume that K is a number field and $a \in K$. We give two equidistribution theorems. Theorem 4.1 will be used in Section 4.4, and Theorem 4.3 will be used in Sections 4.2 and 4.3.

We call $\boldsymbol{E} = \{E_v\}_{v \in M_K}$ an *adelic compact set* if E_v is a non-empty compact subset of $\mathbb{A}_v^{1,\mathrm{an}}$ for any $v \in M_K$ and $E_v = \mathcal{D}(0,1)$ for all but finitely many $v \in M_K$.

Theorem 4.1 (*cf.* [BR10, Theorem 7.52]). *Let K be a number field, and let $\boldsymbol{E}$ be an adelic compact set with*

$$\gamma(\boldsymbol{E}) = \prod_{v \in M_K} \gamma_v(E_v)^{n_v} = 1.$$

Suppose S_n is a sequence of finite, non-empty, $\mathrm{Gal}(\overline{K}/K)$-invariant subsets of $\overline{K}$ such that $|S_n| \to \infty$ and

$$h_{\boldsymbol{E}}(S_n) = \sum_{v \in M_K} n_v \left(\frac{1}{|S_n|} \sum_{z \in S_n} g_{E_v}(z) \right) \longrightarrow 0.$$

Fix a $v \in M_K$ and, for any $n \geq 1$, let $[S_n]$ be the probability measure on $\mathbb{A}_v^{1,\mathrm{an}}$ supported equally on the elements of S_n. Then the sequence of measures $[S_n]$ converges weakly to μ_{E_v} on $\mathbb{A}_v^{1,\mathrm{an}}$.

For related results, see also [Bil97, BR06, CL06, FRL06] and [BR10, Theorem 10.24].

From Theorem 2.1 and Proposition 3.8, we know that $\boldsymbol{M}_a = \{M_{a,v}\}_{v \in M_K}$ is an adelic compact set with $\gamma(\boldsymbol{M}_a) = 1$. Let S_n be the set of all roots of $f_T^n(a)$. The following result shows that $\mu_{a,v}$ can be approximated by $[S_n]$ if we assume that all roots of $f_T^n(a)$ are simple.

Proposition 4.2. *Assume that, for any $n \geq 1$, all roots of $f_T^n(a)$ are simple. Let S_n be the set of all roots of $f_T^n(a)$. Then for any $v \in M_K$, $[S_n]$ converges weakly to $\mu_{a,v}$ on $\mathbb{A}_v^{1,\mathrm{an}}$.*

Proof. By the assumption, we have $|S_n| = d^{n-1} \to \infty$ as $n \to \infty$. To apply Theorem 4.1, it remains to show that $h_{\boldsymbol{M}_a}(S_n) \to 0$ as $n \to \infty$. If $t \in S_n$, then by Theorem 2.1,

$$g_{a,v}(t) = \lim_{k \to \infty} \frac{1}{d^{n+k}} \log^+ \left| f_t^{n+k+1}(a) \right|_v = \frac{1}{d^n} \lim_{k \to \infty} \frac{1}{d^k} \log^+ \left| f_t^{k+1}(0) \right|_v = \frac{1}{d^n} g_{0,v}(t).$$

Since $g_{0,v}$ is continuous, it suffices to show that for any $v \in M_K$ and any $t \in S_n$, $|t|_v$ is bounded by some constant $C_{a,v}$, where $C_{a,v}$ is independent of n.

(1) If $v \in M_K^0$ and $|a|_v \leq 1$, then by Proposition 3.8, we have $|t|_v \leq 1$ for any $t \in S_n$.
(2) If $v \in M_K^0$ and $|a|_v > 1$, then by Proposition 3.10, we have $|t|_v = |a|_v^d$ for any $t \in S_n$.
(3) If $v \in M_K^\infty$, then by Proposition 3.4, the roots of $f_T^n(a) = a$ are inside

$$D_{a,v} = D(0, 3\max\{|a|_v, 4\}^d).$$

Once we can show that $|f_t^n(a)|_v > |a|_v$ for any $t \in \partial D_{a,v}$, we can apply Rouché's theorem to conclude $S_n \subseteq D_{a,v}$. If $t \in \partial D_{a,v}$, then

$$|f_t(a)|_v \geq |t|_v - |a|_v^d = 3\max\{|a|_v, 4\}^d - |a|_v^d \geq 2\max\{|a|_v, 4\}^d \geq 8\max\{|a|_v, 4\}.$$

Assume that $|f_t^n(a)|_v \geq 8\max\{|a|_v, 4\}$ for some $n \geq 1$; then

$$\left| f_t^{n+1}(a) \right|_v \geq |f_t^n(a)|_v^d - |t|_v \geq \left(8^d - 3\right) \max\{|a|_v, 4\}^d \geq 8\max\{|a|_v, 4\}.$$

By induction, $|f_t^n(a)|_v \geq 8\max\{|a|_v, 4\} > |a|_v$ for any $n \geq 1$. □

Proposition 4.10 shows that if $|a|_v > |d|_v^{-2/(d-1)}$ for some $v \in M_K^0$, then all roots of $f_T^n(a)$ are simple. Since we lack a similar result for $v \in M_K^\infty$, we need an equidistribution theorem taking the multiplicity into account.

Theorem 4.3. *Fix a $v \in M_K^\infty$ and a sequence $0 \le k(n) < n$. Let δ_n be the discrete probability measure on $\mathbb{C}$ weighted by multiplicity on the roots of $f_T^n(a) = f_T^{k(n)}(a)$. Then the sequence of measures δ_n converges weakly to $\mu_{a,v}$ on $\mathbb{C}$.*

Proof. See [BD11, Section 4.3] and the proof of [DF08, Theorem 1]. □

4.2. Archimedean estimates for $d = 2$

In this section, we assume that K is a number field, $a, b \in K$, $v \in M_K^\infty$, and $d = 2$. Because v is fixed, we write $|\cdot|$, M_a, μ_a, and g_a for $|\cdot|_v$, $M_{a,v}$, $\mu_{a,v}$, and $g_{a,v}$.

As [DKY22, Section 3.2], we first cover M_a by some disjoint open disks around the roots of $f_T^n(a) = f_T(a)$ with $n = 2, 3$, and then we estimate the lower bound for $g_a(t)$ when t is outside the cover. Let

$$\alpha_1(a) = -a^2 - a, \quad \alpha_2(a) = -a^2 + a, \quad \alpha_3(a) = -a^2 - a - 1, \quad \alpha_4(a) = -a^2 + a - 1$$

be the roots of $f_T^3(a) = f_T(a)$. Note that $\alpha_1(a)$ and $\alpha_2(a)$ are also roots of $f_T^2(a) = f_T(a)$.

Proposition 4.4. *Assume that $|a| \ge 28$. Then we have*

(1) $M_a \subseteq \cup_{i=1}^2 D(\alpha_i(a), 10)$;
(2) *if $t \notin \cup_{i=1}^2 D(\alpha_i(a), 10)$, then*
$$g_a(t) \ge \frac{1}{2}\log(13|a|);$$
(3) $\mu_a(D(\alpha_i(a), 10)) = 1/2$ *for $i = 1, 2$.*

Proof. For simplicity, let $\alpha_i = \alpha_i(a)$ and $D_i = D(\alpha_i, 10)$.

(1) If $t \in \partial D_1$, then

$$|t - \alpha_1| = 10,$$
$$|t - \alpha_2| \ge |\alpha_1 - \alpha_2| - |t - \alpha_1| = 2|a| - 10.$$

When $|a| \ge 28$, we have

$$|f_t^2(a) - f_t(a)| \ge 20|a| - 100 \ge 16|a|.$$

The same reasoning also works for $t \in \partial D_2$. By Rouché's theorem, for any c with $|c| < 16|a|$, the equation $f_T^2(a) - f_T(a) = c$ has exactly one root in each D_i. By Proposition 3.3,

$$\left|f_s^2(a) - f_s(a)\right| \le \left|f_s^2(a)\right| + |f_s(a)| < 16|a|$$

for any $s \in M_a$, so we have $M_a \subseteq \cup_{i=1}^2 D_i$.

(2) If $t \in \partial D_i$ for some $i = 1, 2$, then

$$|t| \le |t - \alpha_i| + |\alpha_i| \le 2|a|^2,$$
$$|f_t(a)| \le |t - \alpha_i| + |\alpha_i + a^2| \le 2|a|$$

and

$$\left|f_t^2(a)\right| \ge \left|f_t^2(a) - f_t(a)\right| - |f_t(a)| \ge 14|a|.$$

Let $p(z) = z^2 - 2$; then by induction and an argument similar to that of Lemma 3.7,

$$\left|f_t^{n+2}(a)\right| \ge p^n(14)|a|^{2^n} \ge (13|a|)^{2^n}$$

for any $n \ge 1$. Therefore, by Theorem 2.1,

$$g_a(t) = \lim_{n\to\infty} \frac{1}{2^{n+1}} \log^+ \left|f_t^{n+2}(a)\right| \ge \frac{1}{2}\log(13|a|).$$

Since g_a is harmonic on $\mathbb{C}\backslash M_a$, this is true for any $t \notin \cup_{i=1}^2 D_i$.

(3) If $t \in \partial D_i$ for some $i = 1, 2$, then

$$|f_t(a) \pm a| \le |f_t(a)| + |a| \le 3|a|$$

and

$$|f_t^n(a) - f_t(a)| \ge |f_t^n(a)| - |f_t(a)| \ge (13|a|)^{2^{n-2}} - 2|a| > 3|a|$$

for any $n \ge 2$. By Rouché's theorem, $f_T^n(a) - f_T(a)$ and $f_T^n(a) \pm a$ have the same number of roots in D_i. Since

$$f_T^{n+1}(a) - f_T(a) = \Big(f_T^n(a) + a\Big)\Big(f_T^n(a) - a\Big),$$

by induction each of $f_T^n(a) - f_T(a)$ and $f_T^n(a) \pm a$ has 2^{n-2} roots in D_i. Then the conclusion follows from Theorem 4.3. □

Proposition 4.5. *Assume that* $|a| \ge 28$. *Then we have*

(1) $M_a \subseteq \cup_{i=1}^4 D(\alpha_i(a), 5/|a|)$;
(2) *if* $t \notin \cup_{i=1}^4 D(\alpha_i(a), 5/|a|)$, *then*

$$g_a(t) \ge \frac{1}{4}\log(13|a|);$$

(3) $\mu_a(D(\alpha_i(a), 5/|a|)) = 1/4$ *for* $1 \le i \le 4$.

Proof. For simplicity, let $\alpha_i = \alpha_i(a)$ and $D_i = D(\alpha_i, 5/|a|)$. If $t \in \partial D_1$, then

$$\begin{aligned}
&|t - \alpha_1| = 5/|a|,\\
&|t - \alpha_2| \ge |\alpha_1 - \alpha_2| - |t - \alpha_1| = 2|a| - 5/|a|,\\
&|t - \alpha_3| \ge |\alpha_1 - \alpha_3| - |t - \alpha_1| = 1 - 5/|a|,\\
&|t - \alpha_4| \ge |\alpha_1 - \alpha_4| - |t - \alpha_1| \ge 2|a| - 1 - 5/|a|.
\end{aligned}$$

When $|a| \ge 28$, we have

$$\left|f_t^3(a) - f_t(a)\right| \ge 20|a| - 110 - \frac{50}{|a|} + \frac{525}{|a|^2} - \frac{625}{|a|^4} \ge 16|a|.$$

The rest of the proof is similar to that of Proposition 4.4. □

As [DKY22, Theorem 4.1], we estimate the complex Arakelov–Zhang pairing as follows.

Proposition 4.6. *For any* $a, b \in K$, *we have*

$$\int g_a d\mu_b \ge \frac{1}{8}\log^+\left|a^2 - b^2\right| - \frac{1}{8}\log 5000.$$

Moreover, if $\max\{|a|, |b|\} \ge 50$ *and*

$$\left|a^2 - b^2\right| \ge \frac{11}{\max\{|a|, |b|\}}, \tag{4.1}$$

then

$$\int g_a d\mu_b \ge \frac{1}{16}\log\max\{|a|, |b|\}.$$

Proof. Without loss of generality, we assume that $|a| \ge |b|$. We prove the second assertion in parts (1), (2), (3), and we prove the first assertion in part (4).

(1) $\boxed{|a| \ge 50 \text{ and } |b| \le 28}$ If $t \in M_b$, then by Proposition 3.4,

$$|t| \le 3\max\{|b|, 4\}^2 \le 3 \cdot 28^2 \le 50(50 - 2) \le |a|(|a| - 2) = |a|^2 - 2|a|$$

and

$$|f_t(a)| \ge |a|^2 - |t| \ge 2|a|.$$

Let $p(z) = z^2 - 1$; then by induction and an argument similar to that of Lemma 3.7,

$$\left|f_t^{n+1}(a)\right| \geq p^n(2)|a|^{2^n} \geq |a|^{2^n}$$

for any $n \geq 1$. Therefore, by Theorem 2.1,

$$g_a(t) = \lim_{n\to\infty} \frac{1}{2^n} \log^+ \left|f_t^{n+1}(a)\right| \geq \log|a|$$

and

$$\int g_a d\mu_b \geq \log|a|.$$

(2) $\boxed{|a| \geq 50,\ |b| \geq 28, \text{ and } |a^2 - b^2| \geq 20}$

Claim. Some $D(\alpha_i(b), 10)$ is disjoint from $\cup_{i=1}^2 D(\alpha_i(a), 10)$.

Suppose the claim is false; then for each i, there exists a k_i such that

$$\left|\alpha_{k_i}(a) - \alpha_i(b)\right| < 20.$$

If $k_1 = k_2$, then

$$2|b| = |\alpha_1(b) - \alpha_2(b)| \leq \sum_{i=1}^{2} |\alpha_{k_i}(a) - \alpha_i(b)| < 40,$$

and we have a contradiction. If $k_1 \neq k_2$, then

$$2\left|a^2 - b^2\right| = \left|\sum_{i=1}^{2} \alpha_i(a) - \sum_{i=1}^{2} \alpha_i(b)\right| \leq \sum_{i=1}^{2} \left|\alpha_{k_i}(a) - \alpha_i(b)\right| < 40,$$

and we also have a contradiction. Therefore, the claim is proved and, by Proposition 4.4,

$$\int g_a d\mu_b \geq \frac{1}{2} \cdot \frac{1}{2} \log(13|a|) = \frac{1}{4} \log(13|a|).$$

(3) $\boxed{|a| \geq 50,\ |b| \geq 28, \text{ and } 11/|a| \leq |a^2 - b^2| \leq 20}$

Claim. Some $D(\alpha_i(b), 5/|b|)$ is disjoint from $\cup_{i=1}^4 D(\alpha_i(a), 5/|a|)$.

Suppose the claim is false; then for each i, there exists a k_i such that

$$\left|\alpha_{k_i}(a) - \alpha_i(b)\right| < 5/|a| + 5/|b|.$$

If $k_i = k_j$ for some $i \neq j$, then

$$1 \leq |\alpha_i(b) - \alpha_j(b)| \leq |\alpha_{k_i}(a) - \alpha_i(b)| + |\alpha_{k_j}(a) - \alpha_j(b)| < 10/|a| + 10/|b|,$$

and we have a contradiction. If $k_i \neq k_j$ for any $i \neq j$, then

$$4\left|a^2 - b^2\right| = \left|\sum_{i=1}^{4} \alpha_i(a) - \sum_{i=1}^{4} \alpha_i(b)\right| \leq \sum_{i=1}^{4} |\alpha_{k_i}(a) - \alpha_i(b)| < 20/|a| + 20/|b|.$$

This also gives a contradiction because

$$\frac{|a|}{|b|} \leq \left(\left|\frac{a^2}{b^2} - 1\right| + 1\right)^{1/2} = \left(\frac{\left|a^2 - b^2\right|}{|b|^2} + 1\right)^{1/2} \leq \left(\frac{20}{28^2} + 1\right)^{1/2} \leq \frac{6}{5}$$

and

$$\left|a^2 - b^2\right| \geq 11/|a| \geq 5/|a| + 5/|b|.$$

Therefore, the claim is proved and, by Proposition 4.5,

$$\int g_a d\mu_b \geq \frac{1}{4} \cdot \frac{1}{4} \log(13|a|) = \frac{1}{16} \log(13|a|).$$

(4) Now we prove the first assertion. If $|a| \le 50$ or $|a^2 - b^2| \le 20$, then

$$\log^+ \left|a^2 - b^2\right| \le \log \max\left\{2|a|^2, 20\right\} \le \log 5000$$

and

$$\int g_a d\mu_b \ge 0 \ge \frac{1}{8}\log^+ \left|a^2 - b^2\right| - \frac{1}{8}\log 5000.$$

If $|a| \ge 50$ and $|a^2 - b^2| \ge 20$, then by parts (1) and (2),

$$\int g_a d\mu_b \ge \frac{1}{4}\log(13|a|) \ge \frac{1}{8}\log\left(2|a|^2\right) \ge \frac{1}{8}\log^+ \left|a^2 - b^2\right|. \qquad \square$$

4.3. Archimedean estimates for $d > 2$

In this section, we assume that K is a number field, $a, b \in K$, $v \in M_K^\infty$, and $d > 2$. Because v is fixed, we write $|\cdot|$, M_a, μ_a, and g_a for $|\cdot|_v$, $M_{a,v}$, $\mu_{a,v}$, and $g_{a,v}$.

We will prove the counterparts of Propositions 4.4, 4.5, and 4.6 for $d > 2$, but this time we only need to consider the roots of $f_T^2(a) = f_T(a)$. The reason is as follows: In Equations (4.1) and (4.2), what we really need is

$$|a^d - b^d| \ge \frac{c_1(d)}{\max\{|a|, |b|\}^{c_2(d)}}$$

for some $c_1(d), c_2(d) > 0$. When we consider the roots of $f_T^2(a) = f_T(a)$, we will get $c_2(d) = d - 2$. This is already good enough for $d > 2$, so we no longer need to consider the roots of $f_T^3(a) = f_T(a)$. For $0 \le i \le d-1$, let

$$\alpha_i(a) = -a^d + \zeta_d^i a$$

be the roots of $f_T^2(a) = f_T(a)$.

Proposition 4.7. *Assume that* $|a| \ge 6$. *Then we have*

(1) $M_a \subseteq \cup_{i=0}^{d-1} D(\alpha_i(a), 12/|a|^{d-2})$;

(2) *if* $t \notin \cup_{i=0}^{d-1} D(\alpha_i(a), 12/|a|^{d-2})$, *then*

$$g_a(t) \ge \frac{1}{d}\log(13|a|);$$

(3) $\mu_a(D(\alpha_i(a), 12/|a|^{d-2})) = 1/d$ *for* $0 \le i \le d-1$.

Proof. For simplicity, let $\alpha_i = \alpha_i(a)$ and $D_i = D(\alpha_i, 12/|a|^{d-2})$. If $t \in \partial D_0$, then

$$|t - \alpha_0| = 12/|a|^{d-2},$$
$$|t - \alpha_i| \ge |\alpha_0 - \alpha_i| - |t - \alpha_0| = \left|1 - \zeta_d^i\right||a| - 12/|a|^{d-2} \ge (1 - 1/d)\left|1 - \zeta_d^i\right||a|,$$

where the last inequality follows from

$$\left(\frac{12d}{\left|1 - \zeta_d^i\right|}\right)^{\frac{1}{d-1}} \le \left(\frac{12d^2}{d\,|1 - \zeta_d|}\right)^{\frac{1}{d-1}} \le \left(\frac{4d^2}{\sqrt{3}}\right)^{\frac{1}{d-1}} \le 2 \cdot 3^{3/4} \le |a|.$$

Then we have

$$|f_t^2(a) - f_t(a)| \ge \frac{12}{|a|^{d-2}} \prod_{i=1}^{d-1} \left(\left(1 - \frac{1}{d}\right)|1 - \zeta_d^i| \cdot |a|\right) = 12d\left(1 - \frac{1}{d}\right)^{d-1} |a| \ge 16|a|.$$

The rest of the proof is similar to that of Proposition 4.4. $\square$

Proposition 4.8. *For any* $a, b \in K$, *we have*

$$\int g_a d\mu_b \ge \frac{1}{d^3}\log^+ \left|a^d - b^d\right| - \frac{1}{d^3}\log\left(2 \cdot 9^d\right).$$

Moreover, if $\max\{|a|,|b|\} \geq 9$ *and*

$$\left|a^d - b^d\right| \geq \frac{25}{\max\{|a|,|b|\}^{d-2}}, \tag{4.2}$$

then

$$\int g_a d\mu_b \geq \frac{1}{d^2} \log \max\{|a|,|b|\}.$$

Proof. Without loss of generality, we assume that $|a| \geq |b|$. We prove the second assertion in parts (1), (2), and we prove the first assertion in part (3).

(1) $\boxed{|a| \geq 9 \text{ and } |b| \leq 6}$ If $t \in M_b$, then by Proposition 3.4,

$$|t| \leq 3\max\{|b|,4\}^d \leq 3 \cdot 6^d \leq 9\left(9^{d-1} - 2\right) \leq |a|\left(|a|^{d-1} - 2\right) = |a|^d - 2|a|$$

and

$$|f_t(a)| \geq |a|^d - |t| \geq 2|a|.$$

By an argument similar to that of the proof of Proposition 4.6, we have

$$\int g_a d\mu_b \geq \log |a|.$$

(2) $\boxed{|a| \geq 9,\ |b| \geq 6, \text{ and } |a^d - b^d| \geq 25/|a|^{d-2}}$

Claim. Some $D(\alpha_i(b), 12/|b|^{d-2})$ is disjoint from $\cup_{i=0}^{d-1} D(\alpha_i(a), 12/|a|^{d-2})$.

Suppose the claim is false; then for each i, there exists a k_i such that

$$|\alpha_{k_i}(a) - \alpha_i(b)| < 12/|a|^{d-2} + 12/|b|^{d-2}.$$

If $k_i = k_j$ for some $i \neq j$, then

$$\begin{aligned}\left|\zeta_d^i - \zeta_d^j\right||b| = |\alpha_i(b) - \alpha_j(b)| &\leq |\alpha_{k_i}(a) - \alpha_i(b)| + |\alpha_{k_j}(a) - \alpha_j(b)| \\ &< 24/|a|^{d-2} + 24/|b|^{d-2} \leq 48/|b|^{d-2}.\end{aligned}$$

This gives a contradiction because

$$\left(\frac{48}{\left|\zeta_d^i - \zeta_d^j\right|}\right)^{\frac{1}{d-1}} \leq \left(\frac{48d}{d\,|1-\zeta_d|}\right)^{\frac{1}{d-1}} \leq \left(\frac{16d}{\sqrt{3}}\right)^{\frac{1}{d-1}} \leq 4 \cdot 3^{1/4} \leq |b|.$$

If $k_i \neq k_j$ for any $i \neq j$, then

$$d|a^d - b^d| = \left|\sum_{i=0}^{d-1} \alpha_i(a) - \sum_{i=0}^{d-1} \alpha_i(b)\right| \leq \sum_{i=0}^{d-1} \left|\alpha_{k_i}(a) - \alpha_i(b)\right| < d\left(12/|a|^{d-2} + 12/|b|^{d-2}\right).$$

If $|a^d - b^d| \geq 4$, then

$$4 \leq \left|a^d - b^d\right| < 12/|a|^{d-2} + 12/|b|^{d-2} \leq 24/|b|^{d-2} \leq 4,$$

which is false. If $25/|a|^{d-2} \leq |a^d - b^d| \leq 4$, then

$$\frac{|a|^{d-2}}{|b|^{d-2}} \leq \frac{|a|^d}{|b|^d} \leq \frac{|a^d - b^d|}{|b|^d} + 1 \leq \frac{4}{6^d} + 1 \leq \frac{13}{12}$$

and

$$25/|a|^{d-2} \leq |a^d - b^d| < 12/|a|^{d-2} + 12/|b|^{d-2} \leq 25/|a|^{d-2},$$

which is also false. Therefore, the claim is proved and, by Proposition 4.7,

$$\int g_a d\mu_b \geq \frac{1}{d}\cdot\frac{1}{d}\log(13|a|) = \frac{1}{d^2}\log(13|a|).$$

(3) Now we prove the first assertion. If $|a| \leq 9$ or $|a^d - b^d| \leq 3$, then

$$\log^+\left|a^d - b^d\right| \leq \log\max\left\{2|a|^d, 3\right\} \leq \log\left(2\cdot 9^d\right)$$

and

$$\int g_a d\mu_b \geq 0 \geq \frac{1}{d^3}\log^+\left|a^d - b^d\right| - \frac{1}{d^3}\log\left(2\cdot 9^d\right).$$

If $|a| \geq 9$ and $|a^d - b^d| \geq 3 \geq 25/|a|^{d-2}$, then by parts (1) and (2),

$$\int g_a d\mu_b \geq \frac{1}{d^2}\log(13|a|) \geq \frac{1}{d^3}\log\left(2|a|^d\right) \geq \frac{1}{d^3}\log^+\left|a^d - b^d\right|. \qquad \square$$

4.4. Non-Archimedean estimates

In this section, we assume that K is a number field, $a, b \in K$, and $v \in M_K^0$. Because v is fixed, we write $|\cdot|$, δ, M_a, μ_a, and g_a for $|\cdot|_v$, δ_v, $M_{a,v}$, $\mu_{a,v}$, and $g_{a,v}$.

As in [DKY22, Sections 5.1 and 6.1], we first study the structure of M_a. More precisely, we show that when a is large enough, M_a can be described with respect to the roots of $f_T^n(a)$.

Proposition 4.9. *Assume that $|a| > |d|^{-2/(d-1)}$. Fix a $t \in \mathbb{C}_v$ such that $|f_t^n(a)| \leq |a|$ for some $n \geq 1$. Let $s_1, \ldots, s_{d^{n-1}}$ be the roots of $f_T^n(a)$ such that*

$$|t - s_1| \leq \cdots \leq |t - s_{d^{n-1}}|.$$

Then we have

$$|t - s_1| = \frac{\left|f_t^n(a)\right|}{\left(|d|\cdot|a|^{d-1}\right)^{n-1}} \leq \frac{|a|}{\left(|d|\cdot|a|^{d-1}\right)^{n-1}} < |t - s_2| \leq \frac{|a|}{\left(|d|\cdot|a|^{d-1}\right)^{n-2}} < |t - s_{d+1}|.$$

Proof. Let $A_n = |a|/(|d|\cdot|a|^{d-1})^{n-1}$ for any $n \geq 1$. By Proposition 3.10, $|f_t^n(a)| \leq |a|$ implies $|f_t^k(a)| = |a|$ for any $1 \leq k \leq n-1$. Thus we can prove the assertion by induction. Because $|t + a^d| = |f_t(a)|$, the statement is true for $n = 1$. Assume the statement is true for some $n \geq 1$, and fix a $t \in \mathbb{C}_v$ such that $|f_t^{n+1}(a)| \leq |a|$. Let $\alpha_1, \ldots, \alpha_{d^{n-1}}$ be the roots of $f_T^n(a)$ with $|t - \alpha_i|$ increasing, and let $\beta_1, \ldots, \beta_{d^n}$ be the roots of $f_T^{n+1}(a)$ with $|t - \beta_i|$ increasing. Then

$$\prod_{i=1}^{d^{n-1}}(t - T - \alpha_i)^d + t - T = f_{t-T}^n(a)^d + t - T = f_{t-T}^{n+1}(a) = \prod_{i=1}^{d^n}(t - T - \beta_i).$$

Let

$$\prod_{i=1}^{d^{n-1}}(t - T - \alpha_i)^d = \sum_{i=0}^{d^n} a_i T^i \text{ and } \prod_{i=1}^{d^n}(t - T - \beta_i) = \sum_{i=0}^{d^n} b_i T^i;$$

then by the assumption $|a| > |d|^{-2/(d-1)}$ and the induction hypothesis $|t-\alpha_1| = A_n < |t-\alpha_2|$,

$$|b_0| = \left|\prod_{i=1}^{d^n}(t-\beta_i)\right| = \left|f_t^{n+1}(a)\right|,$$

$$|b_1| = |a_1 - 1| = \left|\sum_{j=1}^{d^{n-1}} \frac{d\prod_{i=1}^{d^{n-1}}(t-\alpha_i)^d}{t-\alpha_j} - 1\right| = \frac{|d|\left|f_t^n(a)\right|^d}{A_n} = \frac{|d|\cdot|a|^d}{A_n},$$

$$|b_i| = |a_i| \le \frac{\left|f_t^n(a)\right|^d}{A_n^i} = \frac{|a|^d}{A_n^i} \quad \text{for any } 2 \le i \le d-1,$$

$$|b_d| = |a_d| = \frac{\left|f_t^n(a)\right|^d}{A_n^d} = \frac{|a|^d}{A_n^d}.$$

Now we consider the Newton polygon of $f_{t-T}^{n+1}(a)$. By the induction hypothesis,

$$\left(\frac{|b_i|}{|b_d|}\right)^{\frac{1}{d-i}} \le A_n < \left(\frac{|a_d|}{|a_j|}\right)^{\frac{1}{j-d}} = \left(\frac{|b_d|}{|b_j|}\right)^{\frac{1}{j-d}}$$

for any $i < d < j$, so $(d, -\log|b_d|)$ is a vertex of the Newton polygon. By the assumptions,

$$\frac{|b_0|}{|b_1|} = \frac{\left|f_t^{n+1}(a)\right|A_n}{|d|\cdot|a|^d} \le \frac{A_n}{|d|\cdot|a|^{d-1}} < |d|A_n \le \left(\frac{|b_1|}{|b_i|}\right)^{\frac{1}{i-1}}$$

for any $2 \le i \le d$, so $(1, -\log|b_1|)$ is also a vertex of the Newton polygon. Therefore,

$$|t-\beta_1| = \frac{\left|f_t^{n+1}(a)\right|A_n}{|d|\cdot|a|^d} \le \frac{A_n}{|d|\cdot|a|^{d-1}} < |t-\beta_2| \le A_n < |t-\beta_{d+1}|.$$

This completes the inductive step and hence the proof. □

In particular, Proposition 4.9 implies the non-Archimedean versions of Propositions 4.4, 4.5, and 4.7 as follows.

Proposition 4.10. *Assume that $|a| > |d|^{-2/(d-1)}$. For any $n \ge 1$, let S_n be the set of all roots of $f_T^n(a)$. Then*

(1) *all roots of $f_T^n(a)$ are simple; moreover, for any $s_1, s_2 \in S_n$, we have*

$$|s_1 - s_2| > \frac{|a|}{\left(|d|\cdot|a|^{d-1}\right)^{n-1}};$$

(2) *$M_a \subseteq \mathbb{C}_v$;*

(3) *for any $s \in S_n$, we have*

$$\mu_a\left(\mathcal{D}\left(s, \frac{|a|}{\left(|d|\cdot|a|^{d-1}\right)^{n-1}}\right)\right) = \frac{1}{d^{n-1}}.$$

Proof. For any $n \ge 1$, let $A_n = |a|/(|d|\cdot|a|^{d-1})^{n-1}$. For any $s \in S_n$, let

$$T_{n,s} = \{t \in \mathbb{C}_v : |s-t| = A_n\}.$$

(1) Fix any $s \in S_n$, and let $s_1, \dots, s_{d^{n-1}} \in S_n$ be such that $|s - s_i|$ is increasing. By Proposition 4.9, we have $|s-s_1| = 0 < A_n < |s-s_2|$, so s is simple.

Claim 1. For any $s \in S_n$, we have $|S_{n+1} \cap T_{n,s}| = d$.

By Propositions 3.10 and 4.9, we have $S_{n+1} \subseteq \cup_{s\in S_n} T_{n,s}$. Since $|S_{n+1}| = d|S_n|$, it suffices to show that $|S_{n+1} \cap T_{n,s}| \le d$ for any $s \in S_n$. Suppose $\alpha_1, \dots, \alpha_{d+1} \in S_{n+1} \cap T_{n,s}$; then for any $2 \le i \le d+1$, we have

$$|\alpha_1 - \alpha_i| \le \max\{|s-\alpha_1|, |s-\alpha_i|\} = A_n,$$

which contradicts Proposition 4.9.

Claim 2. If $t \in T_{n,s}$ for some $s \in S_n$, then $|f_t^n(a)| = |a|$.

By Claim 1, there exists an $\alpha \in S_{n+1} \cap T_{n,s}$. By Proposition 4.9, for any $s' \in S_n \backslash \{s\}$, we have

$$\begin{aligned} |s-t| &= A_n = |s-\alpha|, \\ |s'-t| &= \max\{|s'-\alpha|, |s-\alpha|, |s-t|\} = |s'-\alpha|. \end{aligned}$$

By Proposition 3.10, we have

$$|f_t^n(a)| = |s-t| \prod_{s'\in S_n\backslash\{s\}} |s'-t| = |s-\alpha| \prod_{s'\in S_n\backslash\{s\}} |s'-\alpha| = |f_\alpha^n(a)| = |a|.$$

(2) Let $T_n = \{t \in \mathbb{C}_v : |f_t^n(a)| = |a|\}$; then by Proposition 3.9,

$$M_a \cap \mathbb{C}_v = \cap_{n=1}^\infty T_n \subseteq \cap_{n=1}^\infty \overline{T_n}.$$

By Proposition 4.9 and Claim 2,

$$\begin{aligned} T_n &= \{t \in \mathbb{C}_v : |s-t| = A_n \text{ for some } s \in S_n\}, \\ \overline{T_n} &= \left\{t \in \mathbb{A}_v^{1,\mathrm{an}} : \delta(s,t) = A_n \text{ for some } s \in S_n\right\}. \end{aligned}$$

If $t \in \cap_{n=1}^\infty \overline{T_n}$, then $\operatorname{diam}(t) \le A_n$ for any $n \ge 1$. Since $A_n \to 0$ as $n \to \infty$, we have $t \in \mathbb{C}_v$. Therefore, $\cap_{n=1}^\infty T_n = \cap_{n=1}^\infty \overline{T_n}$ and $M_a \cap \mathbb{C}_v = \overline{M_a \cap \mathbb{C}_v} = M_a$.

(3) By Claim 1 and induction, for any $s \in S_n$ and any $k \ge n+1$, we have $|S_k \cap T_{n,s}| = d^{k-n}$. Then the conclusion follows from Proposition 4.2. □

Propositions 4.11 and 4.12 can be seen as the non-Archimedean versions of Propositions 4.6 and 4.8.

Proposition 4.11. *Assume that* $\max\{|a|,|b|\} > |d|^{-2/(d-1)}$. *If*

$$|a^d - b^d| > \frac{|d|^{-2(n-1)}}{\max\{|a|,|b|\}^{(d-1)(n-1)-1}}$$

for some $n \ge 1$, *then*

$$\int g_a d\mu_b \ge \frac{1}{d^{2n-2}} \log \max\{|a|,|b|\}.$$

Proof. If $|a| \ne |b|$, then by Propositions 3.8 and 3.9,

$$\int g_a d\mu_b = d \log \max\{|a|,|b|,1\} \ge \frac{1}{d^{2n-2}} \log \max\{|a|,|b|\}.$$

From now on we assume that $|a| = |b|$. For any $n \ge 1$, let $A_n = |a|/(|d| \cdot |a|^{d-1})^{n-1}$, and let $\alpha_1(a), \dots, \alpha_{d^{n-1}}(a)$ be the roots of $f_T^n(a)$. For any $1 \le i \le d^{n-1}$, let $D_i(a) = \overline{D}(\alpha_i(a), A_n)$. Define $\alpha_i(b)$ and $D_i(b)$ similarly.

Claim. Some $D_i(b)$ is disjoint from $\cup_{i=1}^{d^{n-1}} D_i(a)$.

Suppose the claim is false; then for each i, there exists a k_i such that

$$|\alpha_{k_i}(a) - \alpha_i(b)| \le A_n.$$

If $k_i = k_j$ for some $i \ne j$, then

$$|\alpha_i(b) - \alpha_j(b)| \le \max\left\{\left|\alpha_{k_i}(a) - \alpha_i(b)\right|, \left|\alpha_{k_j}(a) - \alpha_j(b)\right|\right\} \le A_n,$$

which contradicts Proposition 4.10. If $k_i \neq k_j$ for any $i \neq j$, then

$$|d|^{n-1}\left|a^d - b^d\right| = \left|\sum_{i=1}^{d^{n-1}} \alpha_i(a) - \sum_{i=1}^{d^{n-1}} \alpha_i(b)\right| \leq \max\{|\alpha_{k_i}(a) - \alpha_i(b)|\}_{i=1}^{d^{n-1}} \leq A_n,$$

which contradicts the assumption

$$\left|a^d - b^d\right| > \frac{|d|^{-2(n-1)}}{|a|^{(d-1)(n-1)-1}} = \frac{|a|}{\left(|d|^2|a|^{d-1}\right)^{n-1}} = \frac{A_n}{|d|^{n-1}}.$$

Let $D_i(b)$ be disjoint from $\cup_{i=1}^{d^{n-1}} D_i(a)$ and $t \in D_i(b)$. By Proposition 3.10,

$$|t| = \max\{|t - \alpha_i(b)|, |\alpha_i(b)|\} = |a|^d.$$

Since $t \notin \cup_{i=1}^{d^{n-1}} D_i(a)$, by Proposition 4.9, we have $|f_t^n(a)| > |a|$. By induction, $|f_t^{n+k}(a)| = |f_t^n(a)|^{d^k}$ for any $k \geq 1$. Therefore, by Theorem 2.1,

$$g_a(t) = \lim_{k\to\infty} \frac{1}{d^{n+k-1}} \log^+\left|f_t^{n+k}(a)\right| = \frac{1}{d^{n-1}} \log|f_t^n(a)| \geq \frac{1}{d^{n-1}} \log|a|.$$

By Proposition 4.10,

$$\int g_a d\mu_b \geq \frac{1}{d^{n-1}} \cdot \frac{1}{d^{n-1}} \log|a| = \frac{1}{d^{2n-2}} \log|a|. \qquad \square$$

Proposition 4.12. *For any $a, b \in K$, we have*

$$\int g_a d\mu_b \geq \frac{1}{d^3} \log^+\left|a^d - b^d\right| - \frac{1}{d^3} \log|d|^{-2d/(d-1)}.$$

Proof. If $\max\{|a|, |b|\} \leq |d|^{-2/(d-1)}$ or $|a^d - b^d| \leq |d|^{-2/(d-1)}$, then

$$\log^+\left|a^d - b^d\right| \leq \log\max\left\{|a|^d, |b|^d, |d|^{-2/(d-1)}\right\} \leq \log|d|^{-2d/(d-1)}$$

and

$$\int g_a d\mu_b \geq 0 \geq \frac{1}{d^3} \log^+\left|a^d - b^d\right| - \frac{1}{d^3} \log|d|^{-2d/(d-1)}.$$

If $\max\{|a|, |b|\} > |d|^{-2/(d-1)}$ and

$$\left|a^d - b^d\right| > |d|^{-2/(d-1)} = \frac{|d|^{-2}}{|d|^{-2(d-2)/(d-1)}} \geq \frac{|d|^{-2}}{\max\{|a|, |b|\}^{d-2}},$$

then by Proposition 4.11,

$$\int g_a d\mu_b \geq \frac{1}{d^2} \log\max\{|a|, |b|\} = \frac{1}{d^3} \log\max\left\{|a|^d, |b|^d\right\} \geq \frac{1}{d^3} \log^+\left|a^d - b^d\right|. \qquad \square$$

4.5. Proofs of Theorems 1.3 and 1.4

Now we are ready to give the proofs of Theorems 1.3 and 1.4. Given Propositions 4.6, 4.8, 4.11, and 4.12, our proofs are almost identical to those given in [DKY22, Sections 7 and 8].

Proof of Theorem 1.3. Let K be a number field such that $a, b \in K$. Define

$$(p_v, q_v, m, n) = \begin{cases} (50, 11, 1, 4) & \text{if } d = 2 \text{ and } v \in M_K^\infty, \\ \left(|2|_v^{-2}, |2|_v^{-4}, 1, 4\right) & \text{if } d = 2 \text{ and } v \in M_K^0, \\ (9, 25, d-2, 2) & \text{if } d > 2 \text{ and } v \in M_K^\infty, \\ \left(|d|_v^{-2/(d-1)}, |d|_v^{-2}, d-2, 2\right) & \text{if } d > 2 \text{ and } v \in M_K^0, \end{cases}$$

and $r_v = \max\{|a|_v, |b|_v, p_v\}$ for any $v \in M_K$. Following [DKY22, Section 8.2], we define

$$M_{\text{help}} = \left\{ v \in M_K : \max\{|a|_v, |b|_v\} > p_v,\ \left|a^d - b^d\right|_v > \frac{q_v}{\max\{|a|_v, |b|_v\}^m} \right\},$$

$$M_{\text{close}} = \left\{ v \in M_K : \max\{|a|_v, |b|_v\} > p_v,\ \left|a^d - b^d\right|_v \le \frac{q_v}{\max\{|a|_v, |b|_v\}^m} \right\},$$

$$M_{\text{bounded}} = \{ v \in M_K : \max\{|a|_v, |b|_v\} \le p_v \}.$$

Then

$$\begin{aligned}
0 &= \sum_{v \in M_K} n_v \log \left|a^d - b^d\right|_v \\
&\le \sum_{v \in M_{\text{close}}} n_v \log \frac{q_v}{r_v^m} + \sum_{v \in M_K^\infty \setminus M_{\text{close}}} n_v \log\left(2 r_v^d\right) + \sum_{v \in M_K^0 \setminus M_{\text{close}}} n_v \log r_v^d \\
&\le \sum_{v \in M_{\text{close}}} n_v \log \frac{q_v}{r_v^m} + \sum_{v \in M_K \setminus M_{\text{close}}} n_v \log\left(q_v r_v^d\right) \\
&= \sum_{v \in M_K} n_v \log q_v - \sum_{v \in M_{\text{close}}} n_v \log r_v^m + \sum_{v \in M_K \setminus M_{\text{close}}} n_v \log r_v^d \\
&= \sum_{v \in M_K} n_v \log q_v - \sum_{v \in M_K} n_v \log r_v^m + \sum_{v \in M_{\text{help}}} n_v \log r_v^{d+m} + \sum_{v \in M_{\text{bounded}}} n_v \log r_v^{d+m} \\
&\le \sum_{v \in M_K} n_v \log q_v^{d+m} - \sum_{v \in M_K} n_v \log r_v^m + \sum_{v \in M_{\text{help}}} n_v \log r_v^{d+m} + \sum_{v \in M_K} n_v \log p_v^{d+m} \\
&\le (d+m) \sum_{v \in M_{\text{help}}} n_v \log r_v - mh(a,b) + (d+m) \sum_{v \in M_K} n_v \log(p_v q_v).
\end{aligned}$$

By Propositions 4.6, 4.8, and 4.11,

$$\begin{aligned}
\left\langle \boldsymbol{\mu}_a, \boldsymbol{\mu}_b \right\rangle &\ge \sum_{v \in M_{\text{help}}} n_v \int g_{a,v}\, d\mu_{b,v} \ge \frac{1}{d^n} \sum_{v \in M_{\text{help}}} n_v \log r_v \\
&\ge \frac{1}{d^n} \left(\frac{m}{d+m} h(a,b) - \sum_{v \in M_K} n_v \log(p_v q_v) \right) \\
&= \begin{cases} \frac{1}{48} h(a,b) - \frac{1}{16} \log 35200 & \text{if } d = 2, \\ \frac{d-2}{2d^2(d-1)} h(a,b) - \frac{1}{d^2} \log(225 d^{2d/(d-1)}) & \text{if } d > 2, \end{cases} \\
&\ge \frac{1}{12 d^2} h(a,b) - 1.
\end{aligned}$$

□

The proof of Theorem 1.4 relies on the following two results. Theorem 4.13 is adapted from [DKY22, Theorem 7.1], and Proposition 4.14 is adapted from a continuity argument used in the proof of [DKY22, Theorem 1.6].

Theorem 4.13. *For any $a, b \in \overline{\mathbb{Q}}$, we have*

$$\left\langle \boldsymbol{\mu}_a, \boldsymbol{\mu}_b \right\rangle \ge \frac{1}{d^3} h\left(a^d - b^d\right) - 2,$$

where h is the logarithmic Weil height on $\overline{\mathbb{Q}}$.

Proof. Let K be a number field such that $a, b \in K$. Define

$$r_v = \begin{cases} 5000 & \text{if } d = 2 \text{ and } v \in M_K^\infty, \\ |2|_v^{-4} & \text{if } d = 2 \text{ and } v \in M_K^0, \\ 2 \cdot 9^d & \text{if } d > 2 \text{ and } v \in M_K^\infty, \\ |d|_v^{-2d/(d-1)} & \text{if } d > 2 \text{ and } v \in M_K^0. \end{cases}$$

By Propositions 4.6, 4.8, and 4.12,

$$\begin{aligned} \left\langle \mu_a, \mu_b \right\rangle &= \sum_{v \in M_K} n_v \int g_{a,v} d\mu_{b,v} \geq \sum_{v \in M_K} n_v \left(\frac{1}{d^3} \log^+ \left| a^d - b^d \right|_v - \frac{1}{d^3} \log r_v \right) \\ &= \frac{1}{d^3} h\left(a^d - b^d\right) - \frac{1}{d^3} \sum_{v \in M_K} n_v \log r_v \\ &= \frac{1}{d^3} h\left(a^d - b^d\right) - \begin{cases} \frac{1}{8} \log 80000 & \text{if } d = 2, \\ \frac{1}{d^3} \log\left(2 \cdot 9^d d^{2d/(d-1)}\right) & \text{if } d > 2, \end{cases} \\ &\geq \frac{1}{d^3} h\left(a^d - b^d\right) - 2. \end{aligned}$$

□

Proposition 4.14. *The complex Arakelov–Zhang pairing $\int g_a d\mu_b$ is a continuous function of $(a, b) \in \mathbb{C}^2$.*

Proof. Let $a_n \to a$ and $b_n \to b$ as $n \to \infty$; then by Proposition 3.4, there exists an $r > 0$ such that for any $n \geq 1$,

$$\{a, b, a_n, b_n\}, M_a, M_b, M_{a_n}, M_{b_n} \subseteq \overline{D}(0, r).$$

By [BH88, Proposition 1.2], $g_a(t)$ is a continuous function of $(a, t) \in \mathbb{C}^2$, so $g_a(t)$ is uniformly continuous on the compact set $\overline{D}(0, r) \times \overline{D}(0, r)$. Therefore, as $n \to \infty$,

$$\int g_a d\mu_b - \int g_{a_n} d\mu_{b_n} = \int (g_a - g_{a_n}) d\mu_b + \int (g_b - g_{b_n}) d\mu_{a_n} \longrightarrow 0.$$

□

Proof of Theorem 1.4. Let K be a number field such that $a, b \in K$. Define

$$(p, q, m, n) = \begin{cases} (50, 11, 1, 4) & \text{if } d = 2, \\ (9, 25, d-2, 2) & \text{if } d > 2. \end{cases}$$

Let N be a large number to be determined later, and let

$$\begin{aligned} M_1 &= \left\{ v \in M_K^\infty : \max\{|a|_v, |b|_v\} \leq N, \ \left| a^d - b^d \right|_v \geq q/N^m \right\}, \\ M_2 &= \left\{ v \in M_K^\infty : \max\{|a|_v, |b|_v\} > N, \ \left| a^d - b^d \right|_v \geq q/N^m \right\}, \\ M_3 &= \left\{ v \in M_K^\infty : \left| a^d - b^d \right|_v < q/N^m \right\}. \end{aligned}$$

Since $\sum_{v \in M_K^\infty} n_v = 1$, we have $\sum_{v \in M_i} n_v \geq 1/3$ for some $1 \leq i \leq 3$.

(1) Assume that $\sum_{v \in M_1} n_v \geq 1/3$. Let

$$R_N = \left\{ (c_1, c_2) \in \mathbb{C}^2 : \max\{|c_1|, |c_2|\} \leq N, \ \left| c_1^d - c_2^d \right| \geq q/N^m \right\}.$$

Since R_N is compact, by [BD11, Lemma 3.4] and Proposition 4.14, we have

$$r_N = \min_{(c_1, c_2) \in R_N} \int g_{c_1} d\mu_{c_2} > 0.$$

Therefore,

$$\left\langle \mu_a, \mu_b \right\rangle \geq \sum_{v \in M_1} n_v \int g_{a,v} d\mu_{b,v} \geq \frac{1}{3} r_N.$$

(2) Assume that $\sum_{v\in M_2} n_v \geq 1/3$. If $v \in M_2$, then

$$|a^d - b^d|_v \geq \frac{q}{N^m} \geq \frac{q}{\max\{|a|_v, |b|_v\}^m}.$$

If $N \geq p$, then by Propositions 4.6 and 4.8,

$$\int g_{a,v} d\mu_{b,v} \geq \frac{1}{d^n} \log \max\{|a|_v, |b|_v\} \geq \frac{1}{d^n} \log N.$$

Therefore,

$$\left\langle \boldsymbol{\mu}_a, \boldsymbol{\mu}_b \right\rangle \geq \sum_{v\in M_2} n_v \int g_{a,v} d\mu_{b,v} \geq \frac{1}{3d^n} \log N.$$

(3) Assume that $\sum_{v\in M_3} n_v \geq 1/3$. By Theorem 4.13,

$$\begin{aligned}
\left\langle \boldsymbol{\mu}_a, \boldsymbol{\mu}_b \right\rangle &\geq \frac{1}{d^3} h\left(a^d - b^d\right) - 2 = \frac{1}{d^3} \sum_{v\in M_K} n_v \log^+ \left|a^d - b^d\right|_v - 2 \\
&\geq \frac{1}{d^3} \sum_{v\in M_K\setminus M_3} n_v \log^+ \left|a^d - b^d\right|_v - 2 \geq \frac{1}{d^3} \sum_{v\in M_K\setminus M_3} n_v \log \left|a^d - b^d\right|_v - 2 \\
&= \frac{1}{d^3} \sum_{v\in M_3} n_v \log \frac{1}{\left|a^d - b^d\right|_v} - 2 \geq \frac{1}{3d^3} \log \frac{N^m}{q} - 2,
\end{aligned}$$

which is positive when $N > q^{1/m} e^{6d^3/m}$. □

5. Proof of Theorem 1.1

Finally, we deduce Theorem 1.1 from Theorems 1.2, 1.3, and 1.4.

Proof of Theorem 1.1.

(1) Let $a, b \in \overline{\mathbb{Q}}$ be such that $a^d \neq b^d$ and $|S_{a,b}| > 0$. By Theorems 1.2, 1.3, and 1.4, there exist real numbers $c_1, c_2, c_3(\varepsilon), c_4, \delta > 0$ such that

$$\max\{c_1 h(a,b) - c_2, \delta\} \leq \left\langle \boldsymbol{\mu}_a, \boldsymbol{\mu}_b \right\rangle \leq \left(\varepsilon + \frac{c_3(\varepsilon)}{|S_{a,b}|}\right)(h(a,b) + c_4).$$

Therefore, we have

$$\max\left\{\frac{c_1 h(a,b) - c_2}{h(a,b) + c_4}, \frac{\delta}{h(a,b) + c_4}\right\} \leq \varepsilon + \frac{c_3(\varepsilon)}{|S_{a,b}|}.$$

When

$$h(a,b) = \frac{c_2 + \delta}{c_1},$$

we have

$$\frac{c_1 h(a,b) - c_2}{h(a,b) + c_4} = \frac{\delta}{h(a,b) + c_4} = \frac{c_1 \delta}{c_1 c_4 + c_2 + \delta}.$$

Since

$$\frac{c_1 h(a,b) - c_2}{h(a,b) + c_4} \text{ is increasing} \quad \text{and} \quad \frac{\delta}{h(a,b) + c_4} \text{ is decreasing}$$

as functions of $h(a,b)$, we have

$$\max\left\{\frac{c_1 h(a,b) - c_2}{h(a,b) + c_4}, \frac{\delta}{h(a,b) + c_4}\right\} \geq \frac{c_1 \delta}{c_1 c_4 + c_2 + \delta}.$$

When ε is small enough, we have

$$|S_{a,b}| \leq \frac{c_3(\varepsilon)}{\frac{c_1\delta}{c_1c_4+c_2+\delta}-\varepsilon}.$$

(2) Let $a \in \mathbb{C}\backslash\overline{\mathbb{Q}}$ and $b \in \mathbb{C}$ be such that $a^d \neq b^d$ and $|S_{a,b}| > 0$. Since each $t \in S_{a,b}$ satisfies $f_t^m(a) = f_t^n(a)$ and $f_t^k(b) = f_t^l(b)$ for some $m > n \geq 0$ and $k > l \geq 0$, the field $\mathbb{Q}(a, b, S_{a,b})$ has transcendence degree 1 over $\mathbb{Q}$. We may view $\mathbb{Q}(a, b, S_{a,b})$ as the function field $K(X)$ of an algebraic curve X defined over a number field K. For all but finitely many $x \in X(\overline{\mathbb{Q}})$, the specializations satisfy $a(x)^d \neq b(x)^d$ and $t_1(x) \neq t_2(x)$ for any $t_1, t_2 \in S_{a,b}$. Therefore, the uniform bound for the algebraic case also works for the complex case. □

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